```
sudo mkdir -p /var/log/cassandra
sudo chown -R `whoami` /var/log/cassandra
sudo mkdir -p /var/lib/cassandra
sudo chown -R `whoami` /var/lib/cassandra
```

For Debian-based Linux distributions like Ubuntu, you can directly install it from the Apache repositories, for which you can add the following lines to 'Software Sources':

```
deb http://www.apache.org/dist/cassandra/debian 12x main
deb-src http://www.apache.org/dist/cassandra/debian 12x main
```

Additionally, there are two GPG keys to be added, as follows:

```
gpg --keyserver pgp.mit.edu --recv-keys F758CE318D77295D
gpg --export --armor F758CE318D77295D | sudo apt-key add -

gpg --keyserver pgp.mit.edu --recv-keys 2B5C1B00
gpg --export --armor 2B5C1B00 | sudo apt-key add -
```

Cassandra can now be installed with the simple *sudo apt-get install cassandra*.

The most basic configuration starts by confirming that all the files and folders have proper permissions. The file *conf/cassandra.yaml* lists all the defaults for other important locations.

Cassandra can now be started by running the following command, which starts the server in the foreground and logs all the output on the terminal:

```
bin/cassandra -f
```

To run it in the background as a daemon, use the same command without the *-f* option. To check if it is running properly, try and access the command-line interface by running the *bin/cassandra-cli* command. If there are still no errors and you are greeted with a prompt, then your installation has been successful. You can also try and run the CQL prompt (*bin/cqlsh*). For detailed instructions on how to set up a cluster, access the official documentation, although it does not cover much beyond some basic configuration. In short, you need to install Cassandra on each node similarly, and specify the earlier node as the seed. An IP interface for Gossip and Thrift are also required to be set up beforehand.

Connecting to Cassandra and using client libraries

Cassandra provides several client APIs to access the database directly from your development environment and a lot of client libraries for many languages like Python, Java, Ruby, Perl, PHP, .NET, C++, Erlang, etc. Even if the required library isn't available, you can use the Thrift interface directly. The Thrift framework supports almost every available language, and allows for cross-platform development by writing simple Thrift files that define the interface between a client and server application.

Cassandra also supports a new interface known as CQL (Cassandra Query Language), which is an SQL clone. It syntactically resembles SQL, besides being more suitable for this data model by allowing for column families and skipping on features like joins, which are not required in the context of a NoSQL data model. CQL is basically a wrapper that provides abstraction to the internal Thrift API, and makes the developer's job much easier by simplifying things. A CQL sample used to create a keyspace looks like what's shown below:

```
CREATE KEYSPACE Test
         WITH replication = {'class': 'SimpleStrategy',
'replication_factor': 3};
```

These commands can be run at the *cqlsh* prompt or using drivers available for languages like Java, Python, PHP, Ruby, etc. (These drivers require Thrift to be installed before use.)

The best part about using Cassandra is that it doesn't miss out on the analytical power provided by Hadoop's MapReduce. Setting this up is pretty easy, as Hadoop can directly be overlaid on top of a Cassandra cluster by having Task Trackers that are local to the Cassandra Analytics cluster, while Hadoop's Name Node and Job Tracker remain separate from the cluster on a different machine. Cassandra provides very good integration to facilitate working with Hadoop MapReduce jobs, and to provide easier access to data stored in Cassandra so that it can be used with other higher-level tools like Pig and Hive. If you are unfamiliar with these terms related to Hadoop, you can read my previous article on the topic, published in an earlier issue of this magazine.

So, we have covered enough about Cassandra for a basic understanding of the important terminology. I have, however, skipped a lot of details that would have been beyond the scope of this article, including the various methods by which we can work with Cassandra, and other theoretical portions like replication strategies, network topology management, security, maintenance and performance tuning. END

References

[1] *DataStax Cassandra Documentation:* http://www.datastax.com/docs
[2] Official Documentation Wiki: *http://wiki.apache.org/cassandra/*
[3] Cassandra CQL 3 Documentation: *http://cassandra.apache.org/doc/cql3/CQL.html#createTableStmt*
[4] O'Reilly's Cassandra: The Definitive Guide by Eben Hewitt

By: Ankit Mathur

The author is a geek with a crush on Java, and also loves flirting with almost anything related to databases and Web technologies. Feel free to poke fun at his articles and direct your feedback to *ankitreloaded@gmail.com*.